HW 1 due 1/28 @ 11:59 PM

(1) (a) $\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \end{array}\right]$ $x_1 + 2x_2 + 3x_3 + 4x_4 = 5$

Free variables

$x_1 = 5 - 2x_2 - 3x_3 - 4x_4$

x_2 is free

x_3 is free

x_4 is free

(b) $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 7 & 1 & -15 \\ 2 & 2 & -8 & 0 & 26 \\ 0 & 3 & 3 & 3 & 9 \end{array}\right]$ $\xrightarrow{r_3 + r_1(-2)}$ $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 7 & 1 & -15 \\ 0 & 0 & -10 & -2 & 22 \\ 0 & 3 & 3 & 3 & 9 \end{array}\right]$ $\xrightarrow{\substack{r_1 + r_2(-1) \\ r_4 + r_2(-3)}}$ $\left[\begin{array}{cccc|c} 1 & 0 & -6 & 0 & 17 \\ 0 & 1 & 7 & 1 & -15 \\ 0 & 0 & -10 & -2 & 22 \\ 0 & 0 & -18 & 0 & 54 \end{array}\right]$

$\xrightarrow{r_3 \times (-\frac{1}{5})}$ $\left[\begin{array}{cccc|c} 1 & 0 & -6 & 0 & 17 \\ 0 & 1 & 7 & 1 & -15 \\ 0 & 0 & 1 & 1/5 & -11/5 \\ 0 & 0 & -18 & 0 & 54 \end{array}\right]$ $\xrightarrow{\substack{r_1 + r_3(6) \\ r_2 + r_3(-7)}}$ $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 6/5 & 19/5 \\ 0 & 1 & 0 & -2/5 & 2/5 \\ 0 & 0 & 1 & 1/5 & -11/5 \\ 0 & 0 & -18 & 0 & 54 \end{array}\right]$ $\xrightarrow{r_4 + r_3(18)}$ $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 6/5 & 19/5 \\ 0 & 1 & 0 & -2/5 & 2/5 \\ 0 & 0 & 1 & 1/5 & -11/5 \\ 0 & 0 & 0 & 18/5 & 72/5 \end{array}\right]$

$\xrightarrow{r_4 \times (\frac{5}{18})}$ $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 6/5 & 19/5 \\ 0 & 1 & 0 & -2/5 & 2/5 \\ 0 & 0 & 1 & 1/5 & -11/5 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right]$ $\xrightarrow{\substack{r_1 + r_4(\frac{-6}{5}) \\ r_2 + r_4(\frac{2}{5})}}$ $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1/5 & -11/5 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right]$ $\xrightarrow{r_3 + r_4(\frac{-1}{5})}$

$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right]$

$a = -1$

$b = 2$

$c = -3$

$d = 4$

(c) $\left[\begin{array}{cc|c} \cos\theta & -\sin\theta & u \\ \sin\theta & \cos\theta & v \end{array}\right] \xrightarrow{r_1\left(\frac{1}{\cos\theta}\right)} \left[\begin{array}{cc|c} 1 & -\tan\theta & \frac{u}{\cos\theta} \\ \sin\theta & \cos\theta & v \end{array}\right] \xrightarrow{r_2 + r_1(-\sin\theta)}$

$\left[\begin{array}{cc|c} 1 & -\tan\theta & \frac{u}{\cos\theta} \\ 0 & \cos\theta + (\tan\theta \cdot \sin\theta) & v - u\tan\theta \end{array}\right] = \left[\begin{array}{cc|c} 1 & -\tan\theta & \frac{u}{\cos\theta} \\ 0 & \cos\theta + \frac{\sin^2\theta}{\cos\theta} & v - u\tan\theta \end{array}\right] =$

$\left[\begin{array}{cc|c} 1 & -\tan\theta & \frac{u}{\cos\theta} \\ 0 & \sec\theta & v - u\tan\theta \end{array}\right] \xrightarrow{r_2(\cos\theta)} \left[\begin{array}{cc|c} 1 & -\tan\theta & u\sec\theta \\ 0 & 1 & v\cos\theta - u\sin\theta \end{array}\right]$

$\xrightarrow{r_1 + r_2(\tan\theta)} \left[\begin{array}{cc|c} 1 & 0 & u\sec\theta + v\sin\theta - u\tan\theta\sin\theta \\ 0 & 1 & v\cos\theta - u\sin\theta \end{array}\right]$

$$\begin{cases} x = u\sec\theta + v\sin\theta - u\tan\theta\sin\theta \\ y = v\cos\theta - u\sin\theta \end{cases}$$

$$v\sin\theta + \frac{u}{\cos\theta} - \frac{v\sin^2\theta}{\cos\theta} \rightarrow v\sin\theta + \frac{u(1-\sin^2\theta)}{\cos\theta}$$

$$v\sin\theta + u\cos\theta$$

$$\begin{cases} x = v\sin\theta + u\cos\theta \\ y = v\cos\theta - u\sin\theta \end{cases}$$

(2) (a) $\frac{1}{x^4(x+1)} = \frac{a_1}{x} + \frac{a_2}{x^2} + \frac{a_3}{x^3} + \frac{a_4}{x^4} + \frac{a_5}{x+1}$

$$RHS = \frac{a_1x^3(x+1) + a_2x^2(x+1) + a_3x(x+1) + a_4(x+1) + a_5x^4}{x^4(x+1)}$$

$$= \frac{a_1x^4 + a_1x^3 + a_2x^3 + a_2x^2 + a_3x^2 + a_3x + a_4x + a_4 + a_5x^4}{x^4(x+1)}$$

$$= \frac{(a_1+a_5)x^4 + (a_1+a_2)x^3 + (a_2+a_3)x^2 + (a_3+a_4)x + a_4}{x^4(x+1)}$$

$$\begin{cases} a_1+a_5 = 0 \\ a_1+a_2 = 0 \\ a_2+a_3 = 0 \\ a_3+a_4 = 0 \\ a_4 = 1 \end{cases} \qquad \left[\begin{array}{ccccc|c} 1&0&0&0&1&0 \\ 1&1&0&0&0&0 \\ 0&1&1&0&0&0 \\ 0&0&1&1&0&0 \\ 0&0&0&1&0&1 \end{array}\right] \xrightarrow{r_2-r_1}$$

$$\left[\begin{array}{ccccc|c} 1&0&0&0&1&0 \\ 0&1&0&0&-1&0 \\ 0&1&1&0&0&0 \\ 0&0&1&1&0&0 \\ 0&0&0&1&0&1 \end{array}\right] \xrightarrow[r_5 \leftrightarrow r_4]{r_3-r_2} \left[\begin{array}{ccccc|c} 1&0&0&0&1&0 \\ 0&1&0&0&-1&0 \\ 0&0&1&0&1&0 \\ 0&0&0&1&0&1 \\ 0&0&1&1&0&0 \end{array}\right] \xrightarrow{r_5-r_3}$$

$$\left[\begin{array}{ccccc|c} 1&0&0&0&1&0 \\ 0&1&0&0&-1&0 \\ 0&0&1&0&1&0 \\ 0&0&0&1&0&1 \\ 0&0&1&1&0&0 \end{array}\right] \xrightarrow{r_5-r_4} \left[\begin{array}{ccccc|c} 1&0&0&0&1&0 \\ 0&1&0&0&-1&0 \\ 0&0&1&0&1&0 \\ 0&0&0&1&0&1 \\ 0&0&0&0&-1&-1 \end{array}\right] \xrightarrow[r_2+r_5(-1)]{r_1+r_5}$$

(cont) on next pg

$$\left[\begin{array}{ccccc|c} 1&0&0&0&0&-1 \\ 0&1&0&0&0&1 \\ 0&0&1&0&1&0 \\ 0&0&0&1&0&1 \\ 0&0&0&0&-1&-1 \end{array}\right] \xrightarrow{r_3+r_5} \xrightarrow{r_5\cdot(-1)} \left[\begin{array}{ccccc|c} 1&0&0&0&0&-1 \\ 0&1&0&0&0&1 \\ 0&0&1&0&0&-1 \\ 0&0&0&1&0&1 \\ 0&0&0&0&1&1 \end{array}\right]$$

$$\begin{cases} a_1 = -1 \\ a_2 = 1 \\ a_3 = -1 \\ a_4 = 1 \\ a_5 = 1 \end{cases}$$

2) a) cont.) $\int \frac{1}{x^4(x+1)}\,dx = \int -\frac{1}{x}\,dx + \int \frac{1}{x^2}\,dx + \int \frac{-1}{x^3}\,dx + \int \frac{1}{x^4}\,dx + \int \frac{1}{x+1}\,dx =$

$-\ln|x| - \frac{1}{x} + \frac{1}{2x^2} - \frac{1}{3x^3} + \ln|x+1| + C$

(2) (b)

$$\text{RHS} = a_1(x-1)(x^2+1) + a_2(x)(x^2+1) + a_3(x)(x-1) + a_4x^2(x-1)$$

$$= a_1x^3 - a_1x^2 + a_1x - a_1 + a_2x^3 + a_2x + a_3x^2 - a_3x + a_4x^3 - a_4x^2$$

$$= (a_1 + a_2 + a_4)x^3 + (-a_1 + a_3 - a_4)x^2 + (a_1 + a_2 - a_3)x - a_1$$

①
$$\begin{cases} a_1 + a_2 + a_4 = 0 \\ -a_1 + a_3 - a_4 = 0 \\ a_1 + a_2 - a_3 = 0 \\ -a_1 = 1 \end{cases}$$

② $\left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 0 \\ -1 & 0 & 1 & -1 & 0 \\ 1 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[r_3+r_1]{r_2+r_1} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ -1 & 0 & 0 & 0 & 1 \end{array}\right]$

$\xrightarrow{r_4+r_1}$ $\left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array}\right] \xrightarrow[r_3+r_2(-1)]{r_1+r_2(-1)} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array}\right] \xrightarrow{r_4+r_2(-1)}$

$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right] \xrightarrow{r_3+r_4(-3)} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right] \xrightarrow[r_2+r_3(-1)]{r_1+r_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & -2 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right]$

$\xrightarrow{r_4+r_3}$ $\left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & -2 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -2 & -1 \end{array}\right] \xrightarrow{r_4 \times (-\frac{1}{2})} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & -2 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 1 & 1/2 \end{array}\right] \xrightarrow[r_2+r_4(-3)]{r_1+r_4(2)} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 1 & 1/2 \end{array}\right]$

$\xrightarrow{r_3+r_4(3)}$ $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 0 & 1/2 \\ 0 & 0 & 0 & 1 & 1/2 \end{array}\right]$ $\begin{cases} a_1 = -1 \\ a_2 = 1/2 \\ a_3 = 1/2 \\ a_4 = 1/2 \end{cases}$

③ $\dfrac{1}{x(x-1)(x^2+1)} = \dfrac{-1}{x} + \dfrac{1}{2(x-1)} + \dfrac{1}{2(x^2+1)} + \dfrac{x}{2(x^2+1)}$

④ $\displaystyle\int \left(\frac{-1}{x} + \frac{1}{2(x-1)} + \frac{1}{2(x^2+1)} + \frac{x}{2(x^2+1)}\right) dx =$

$$-\ln|x| + \frac{1}{2}\ln|x-1| + \frac{1}{2}\arctan x + \frac{1}{4}\ln(x^2+1) + C$$

$u = x^2+1$

$\frac{du}{dx} = 2x$ $\quad du = 2x\,dx$ $\quad \frac{du}{2x} = dx$

$$\frac{1}{2}\int \frac{x}{x^2+1}\,dx = \frac{1}{2}\int \frac{1}{2u}\,du = \frac{1}{4}\int \frac{1}{u}\,du = \frac{1}{4}\ln|u| + C = \frac{1}{4}\ln|x^2+1| + C$$

(3) $x_1Cu + x_2HNO_3 \longrightarrow x_3Cu(NO_3)_2 + x_4NO + x_5H_2O$

Cu: $x_1 = x_3$
H: $x_2 = 2x_5$
N: $x_2 = 2x_3 + x_4$
O: $3x_2 = 6x_3 + x_4 + x_5$

$$\begin{cases} x_1 - x_3 = 0 \\ x_2 - 2x_5 = 0 \\ x_2 - 2x_3 - x_4 = 0 \\ 3x_2 - 6x_3 - x_4 - x_5 = 0 \end{cases}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 1 & -2 & -1 & 0 & 0 \\ 0 & 3 & -6 & -1 & -1 & 0 \end{array}\right] \xrightarrow{r_3 - r_2} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & -2 & -1 & 2 & 0 \\ 0 & 3 & -6 & -1 & -1 & 0 \end{array}\right] \xrightarrow{r_4 - r_2(3)} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & -2 & -1 & 2 & 0 \\ 0 & 0 & -6 & -1 & 5 & 0 \end{array}\right]$$

$$\xrightarrow{r_3 \cdot (-\frac{1}{2})} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -1 & 0 \\ 0 & 0 & -6 & -1 & 5 & 0 \end{array}\right] \xrightarrow{r_4 + r_3} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -1 & 0 \\ 0 & 0 & 0 & -2 & 4 & 0 \end{array}\right] \xrightarrow{r_1 + r_3} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & \frac{1}{2} & -1 & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -1 & 0 \\ 0 & 0 & 0 & -2 & 4 & 0 \end{array}\right]$$

$$\xrightarrow{r_4(-\frac{1}{2})} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & \frac{1}{2} & -1 & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 0 \end{array}\right] \xrightarrow[r_3 + r_4(-\frac{1}{2})]{r_1 + r_4(-\frac{1}{2})} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -\frac{3}{4} & 0 \\ 0 & 1 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & -\frac{3}{4} & 0 \\ 0 & 0 & 0 & 1 & -2 & 0 \end{array}\right] \begin{cases} x_1 - \frac{3}{4}x_5 = 0 \\ x_2 - 2x_5 = 0 \\ x_3 - \frac{3}{4}x_5 = 0 \\ x_4 - \frac{1}{2}x_5 = 0 \\ x_5 \text{ is free} \end{cases}$$

$x_5 = s, s \in \mathbb{R}$

$$\begin{cases} x_1 - \frac{3}{4}s = 0 \\ x_2 - 2s = 0 \\ x_3 - \frac{3}{4}s = 0 \\ x_4 - \frac{1}{2}s = 0 \\ x_5 \text{ free} \end{cases} = \begin{cases} x_1 = \frac{3}{4}s \\ x_2 = 2s \\ x_3 = \frac{3}{4}s \\ x_4 = \frac{1}{2}s \\ x_5 = s \end{cases}$$

Using LCD, $s = 4$ to get integers:

$$= \begin{cases} x_1 = 3 \\ x_2 = 8 \\ x_3 = 3 \\ x_4 = 2 \\ x_5 = 4 \end{cases}$$

$$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

(4) (a) $\underset{2\times2}{\begin{bmatrix} 8 & 1 \\ 7 & 2 \end{bmatrix}} \underset{2\times1}{\begin{bmatrix} 2 \\ -2 \end{bmatrix}} = \begin{bmatrix} 8\times2+1\times-2 \\ 7\times2+2\times-2 \end{bmatrix} = \begin{bmatrix} 16-2 \\ 14-4 \end{bmatrix} = \underset{2\times1}{\begin{bmatrix} 14 \\ 10 \end{bmatrix}}$

(b) $\underset{2\times3}{\begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & -1 \end{bmatrix}} \underset{2\times2}{\begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}}$ Because n in m×n and n×p are not equal. (3 ≠ 2), you cannot multiply a product matrix. It is not defined.

(c) $\underset{3\times2}{\begin{bmatrix} 2 & 3 \\ 1 & 1 \\ 2 & -1 \end{bmatrix}} \underset{2\times2}{\begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}} = \begin{bmatrix} 2\times2+3\times3 & 2\times1+3\times1 \\ 1\times2+1\times3 & 1\times1+1\times1 \\ 2\times2+-1\times3 & 2\times1+-1\times1 \end{bmatrix} = \begin{bmatrix} 4+9 & 2+3 \\ 2+3 & 1+1 \\ 4-3 & 2-1 \end{bmatrix} =$

$= \begin{bmatrix} 13 & 5 \\ 5 & 2 \\ 1 & 1 \end{bmatrix}$

(5) (a) $T(x) = \begin{bmatrix} x \\ x^2 \\ x^3 \end{bmatrix}$ Let $\vec{v} = 2$, $\vec{w} = 1$ NO

$T(\vec{v}+\vec{w}) = T(2+1) = T(3) = \begin{bmatrix} 3 \\ 9 \\ 27 \end{bmatrix}$

$T(\vec{v}) + T(\vec{w}) = T(2) + T(1) = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 9 \end{bmatrix}$

$T(\vec{v}+\vec{w}) \neq T(\vec{v})+T(\vec{w})$, so T is not a linear transformation.

(b) YES $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\vec{w} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $\vec{v}+\vec{w} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ $T(\vec{v}+\vec{w}) = \begin{bmatrix} 3 \\ -25 \end{bmatrix}$

$T(\vec{v}) = \begin{bmatrix} 1 \\ -7 \end{bmatrix}$ $T(\vec{w}) = \begin{bmatrix} 2 \\ -17 \end{bmatrix}$ $T(\vec{v}) + T(\vec{w}) = \begin{bmatrix} 3 \\ -25 \end{bmatrix}$

$T(2\vec{v}) = T\begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -16 \end{bmatrix}$ $2T(\vec{v}) = 2\begin{bmatrix} 1 \\ -8 \end{bmatrix} = \begin{bmatrix} 2 \\ -16 \end{bmatrix}$

✓ $T(\vec{v}) + T(\vec{w}) = T(\vec{v}+\vec{w})$ ✓ $2T(\vec{v}) = T(2\vec{v})$

(5)(b) cont.

$T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\-10\end{bmatrix}$ $\vec{e_1}$ $\quad T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$ $\vec{e_2}$

$A = \begin{bmatrix}1 & 0\\-10 & 1\end{bmatrix}$ checking work: $\underset{2\times2}{\begin{bmatrix}1 & 0\\-10 & 1\end{bmatrix}}\underset{2\times1}{\begin{bmatrix}x\\y\end{bmatrix}} = \underset{2\times1}{\begin{bmatrix}x\\-10x+y\end{bmatrix}}$ ✓

(c) $\vec{v} = \begin{bmatrix}1\\2\\3\end{bmatrix}$ $\vec{w} = \begin{bmatrix}2\\4\\6\end{bmatrix}$

$T(\vec{v}+\vec{w}) = T\begin{bmatrix}3\\6\\9\end{bmatrix} = \begin{bmatrix}3\\-7\end{bmatrix}$ $\quad T(\vec{v})+T(\vec{w}) = T(\vec{v}+\vec{w})$ ✓

$T(\vec{v}) = \begin{bmatrix}1\\-8\end{bmatrix}$ $\quad T(\vec{w}) = \begin{bmatrix}2\\-4\end{bmatrix}$ $\quad T(\vec{w})+T(\vec{w}) \neq \begin{bmatrix}3\\1\end{bmatrix}$

$2\vec{v} = \begin{bmatrix}2\\4\\6\end{bmatrix}$ $\quad T(2\vec{v}) = \begin{bmatrix}2\\-16\end{bmatrix}$ $\quad T(2(\vec{v})) = 2T(\vec{v})$ ✓

$2\,T(\vec{v}) = 2\begin{bmatrix}1\\-8\end{bmatrix} = \begin{bmatrix}2\\-16\end{bmatrix}$

yes, T is a linear transformation

$T\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$ $\quad T\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$ $\quad T\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$

$A = \begin{bmatrix}1 & 0 & 0\\0 & 0 & 1\end{bmatrix}$ checking work: $\underset{2\times3}{\begin{bmatrix}1 & 0 & 0\\0 & 0 & 1\end{bmatrix}}\underset{3\times1}{\begin{bmatrix}x\\y\\z\end{bmatrix}} = \underset{2\times1}{\begin{bmatrix}x\\z\end{bmatrix}}$ ✓

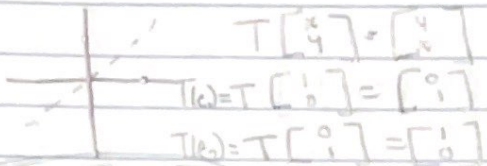
(d) $\vec{v} = 2$ $\quad \vec{w} = 3$ $\quad T(\vec{v}) = 3$ $\quad T(\vec{w}) = 4$ $\quad T(\vec{v})+T(\vec{w}) = 7$

$T(\vec{v}+\vec{w}) = T(5) = 6$

$T(\vec{v}+\vec{w}) \neq T(\vec{v})+T(\vec{w})$ b/c $6 \neq 7$

NO, T is not a linear transformation.

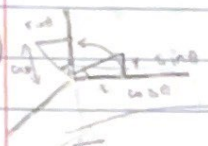
6) a)



$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

checking work: $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$ ✓

b)



z is unchanged

$\cos\theta \to -\sin\theta$

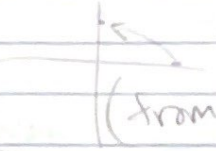
$\sin\theta \to \cos\theta$

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -y \\ x \\ z \end{bmatrix}$$

$T(\vec{e}_1) = T\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$T(\vec{e}_2) = T\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$

$T(\vec{e}_3) = T\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$



(from class) + z

$$\begin{bmatrix} \cos(\pi/2) & -\sin(\pi/2) & 0 \\ \sin(\pi/2) & \cos(\pi/2) & 0 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

7) a) affine transformation:

$T(\vec{x}) = A\vec{x} + \vec{b}$

$T(\vec{x} - \vec{y}) - T(\vec{u} - \vec{v}) = (T(\vec{x}) - T(\vec{y})) - (T(\vec{v}) - T(\vec{u}))$

- substitute using

$[A(\vec{x} - \vec{y}) + \vec{b}] - [A(\vec{u} - \vec{v}) + \vec{b}] = A\vec{x} + \vec{b} - A\vec{y} - \vec{b} - A\vec{u} - \vec{b} + A\vec{v} + \vec{b}$

$A(\vec{x} - \vec{y}) - A(\vec{u} - \vec{v}) = A\vec{x} - A\vec{y} - A\vec{u} + A\vec{v}$

$A\vec{x} - A\vec{y} - A\vec{u} + A\vec{v} = A(\vec{x} - \vec{y} - \vec{u} + \vec{v})$

$A(\vec{x} - \vec{y} - \vec{u} + \vec{v}) = A(\vec{x} - \vec{y} - \vec{u} + \vec{v})$ ✓

- Both sides are in fact equal, so, if T is an affine transformation, $T(\vec{x} - \vec{y}) - T(\vec{u} - \vec{v}) = (T(\vec{x}) - T(\vec{y})) - (T(\vec{u}) - T(\vec{v}))$

b) $T\begin{bmatrix} x \\ y \end{bmatrix} = xy$ ①

② $\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\vec{y} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ $\vec{u} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ $\vec{v} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$

$\vec{x} - \vec{y} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$ $\vec{u} - \vec{v} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

③ $T(\vec{x} - \vec{y}) - T(\vec{u} - \vec{v}) \stackrel{?}{=} (T(\vec{x}) - T(\vec{y})) - (T(\vec{u}) - T(\vec{v}))$

④ $T\begin{bmatrix} -2 \\ -2 \end{bmatrix} - T\begin{bmatrix} -2 \\ -2 \end{bmatrix} \stackrel{?}{=} \left(T\begin{bmatrix} 1 \\ 2 \end{bmatrix} - T\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) - \left(T\begin{bmatrix} 5 \\ 6 \end{bmatrix} - T\begin{bmatrix} 7 \\ 8 \end{bmatrix}\right)$

$4 - 4 \stackrel{?}{=} (2 - 12) - (30 - 56)$

$0 \stackrel{?}{=} (-10) - (-26)$

$0 \neq 16$

NO, T is not an affines transformation.

① 2×2 2×1 2×1

c) $A\begin{bmatrix} 0 \\ 0 \end{bmatrix} + \vec{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $A\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \vec{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ $A\begin{bmatrix} 0 \\ 1 \end{bmatrix} + \vec{b} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$

e) $\vec{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ ③ $A\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ $A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$

$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$A = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$

check work: $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

$\begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} \stackrel{✓}{=} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

2) a) $\left[\begin{array}{cc|cc} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right]$

$A^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ As proved in class, $AA^{-1} = I$

$A = A^{-1}$

$A^{2024} = (AA)^{1012} = (A \cdot A)^{1012} = (A \cdot A^{-1})^{1012} = (I_n)^{1012} = I_n$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

b) $A = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{r_3 - r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 & 0 & 1 \end{array}\right] \xrightarrow{r_3 \cdot \left(\frac{1}{2}\right)}$

$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & 0 & -\frac{1}{2} \end{array}\right] \xrightarrow{r_1 - r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & 0 & -\frac{1}{2} \end{array}\right]$

$A^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} \end{bmatrix}$

$\vec{x} = A^{-1}\vec{b}$

$\vec{x} = \underset{3\times3}{\begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} \end{bmatrix}} \underset{3\times1}{\begin{bmatrix} \pi \\ e \\ 0 \end{bmatrix}} = \underset{3\times1}{\begin{bmatrix} \frac{\pi}{2} + 0 + 0 \\ 0 + e + 0 \\ \frac{\pi}{2} + 0 + 0 \end{bmatrix}} = \begin{bmatrix} \frac{\pi}{2} \\ e \\ \frac{\pi}{2} \end{bmatrix}$